MIL-P-50286A(AR)
6 June 1977
SUPERSEDING
MIL-P-50286(MU)
26 August 1969

MILITARY SPECIFICATION

PROJECTILE, 155 MILLIMETER: M454, WITH ACCESSORIES
DOD COMPONENTS ONLY: ASSEMBLY OF

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers special requirements, provides the quality assurance provisions for the assembling, and the preparation for delivery criterion for one type of Projectile used with 155mm Howitzer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND
PRODUCT AND PACKAGING DRAWINGS

8857468 - Body Ogive Assembly

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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8857469 - Firing Set and Timer, Sequential M32E1 with Plug, Test Assembly
8857477 - Insert, Screw Thread
8857487 - Body-Ogive Matched Machined
8857555 - Nuclear Body Assembly MC-1490
8881785 - Insert Pin
9233664 - Projectile Assembly 155 Millimeter: Shipping Configuration
9233668 - Projectile Assembly, 155 Millimeter: Shipping Configuration
9241967 - Container Shipping and Storage: M467 Pkg and Mkg for Projectile M454A4 or A6.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The projectile assembly shall comply with all requirements specified on Drawing (dwg) 9233664 or 9233668 as applicable with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Interference fit of inserts.

3.4.1 Damaged hole rework. In the event that any insert hole is damaged during the assembly process, the holes may be enlarged in accordance with dwg 8857487. Holes shall be uniform in diameter within .0003 inch for the full depth. All other dimensions shall comply with dwg. 8857487. Oversize mating inserts shall comply with dwg requirements (except diameter) (see dwg 8857477 or dwg 8881785). Each reworked hole shall be identified by applying a number adjacent to the hole using the procedure described on dwg 8857487. Inspection records shall show identification symbol(s) and actual hole and pin diameter(s).

3.4.2 Insert replacement. In the event any insert is extracted from an assembly, the insert hole diameters shall be inspected before insert reassembly to assure that an interference fit of .0008 to .0012 inch will be maintained.

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When inspection results indicate that the proper interference fit will not be obtained at reassembly, the requirements of 3.4.1 for hole rework and/or oversize inserts shall apply.

3.5 Scratches, nicks or gouges. If the scratch, nick or gouge is located near fillets, chamfer, threads, or radii, so that repairing scratch, nick, or gouge will touch the fillet, chamfer, thread, or radii, and is greater than .005 inch deep on the base, rear body and forward body or is greater than .003 inch deep on the center body, the component shall be rejected. If the depth of the scratch, nick, or gouge is .005 inch or less on the base, rear body and forward body or .003 inch or less on the center body when located near a fillet, chamfer, thread, or radii, the scratch, nick, or gouge shall be repaired as specified in 4.4.3. The maximum depth of the repaired imperfection shall not exceed the maximum depth allowed for the imperfection. If the scratch, nick, or gouge in any area other than previously defined on the base and body is greater than .008 inch deep, the component shall be rejected. If the imperfection is .008 inch or less deep, the imperfection shall be repaired as specified in 4.4.3. On the center body the maximum width of the repaired area shall not exceed 0.1 inch and the accumulative length of repairs shall not exceed 20 inches. Maximum depth of any scratch, nick, or gouge repaired not located near a fillet, chamfer, thread, or radii shall not exceed .008 inch.

3.6 Damaged rotating bands. Each rotating band on each projectile assembly shall be inspected for the following defects:

a. Cracks

b. Any dent, nick, scratch or gouge which exceeds the allowable depth (see Figure 1).

c. Foreign matter embedded in rotating band material.

Any rotating band which has any of the defects listed in a, b, and c shall be rejected and removed from the lot when inspected as specified in 4.4.4.

3.7 Serialization. A serial number shall be assigned to each assembly by the Energy Research and Development Administration (ERDA). In no event shall there be any possibility of duplication nor any indication of order or size of production in the serial number or any other number. The serial number assigned shall be coordinated with the Department of Defense (DOD) serial number.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage,

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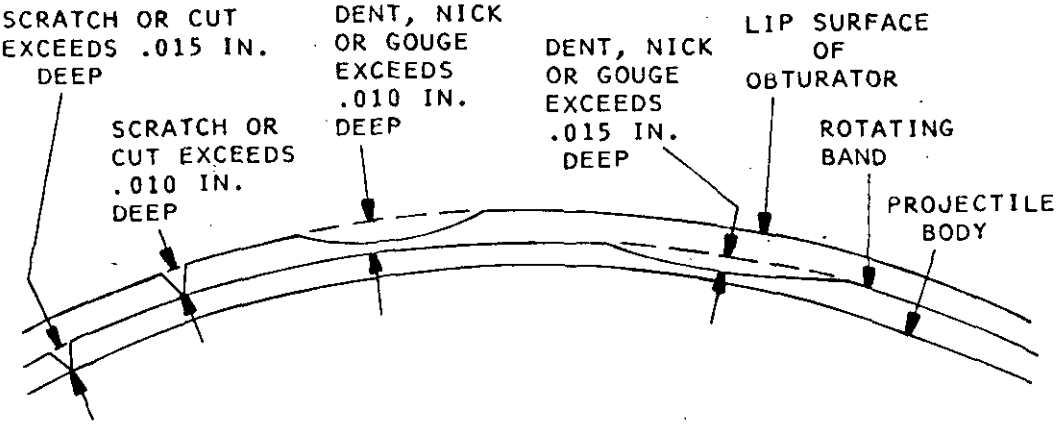


FIGURE 1

ROTATING BAND INSPECTION CRITERIA

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cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg 9233664 or 9233668) for first article testing.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in this specification. A DOD representative shall witness the complete assembly to assure compliance with this specification. In addition the DOD representative will verify that the ERDA inspection criteria and specifications include all requirements, inspections and tests specified in this specification.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with this specification, the first article sample shall be rejected. If the ERDA specifications and inspection criteria do not

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include all requirements, inspections and tests, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table I

Lot size	Sample size 100% inspection
1- 155	155
156- 200	175
201- 250	190
251- 300	200
301- 350	210
351- 400	220
401- 450	225
451- 500	230
501- 550	235
551- 600	240
601- 650	243
651- 700	247
701- 750	250
751- 800	254
801- 900	258
901-1000	261
1001-1100	264
1101-1200	266
1201-1300	268
1301-1400	270
1401-1500	272
1501-1600	

Minor characteristic table II

Lot size	Sample size 100% inspection
1- 25	25
26- 30	26
31- 35	27
36- 40	28
41- 45	29
46- 50	30
51- 55	31
56- 60	32
61- 65	33
66- 70	34
71- 75	35
76- 85	36
86- 95	37
96- 110	38
111- 130	39
131- 156	40
157- 190	41
191- 240	42
241- 340	43
341- 520	44
521-1120	45
1121 and over	

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4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8857555
4.3.2.1.1	Nuclear Body Assembly				NEXT HIGHER ASSEMBLY 8857468
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Depth of insert assemblies		Table I		Manual/Gage
102.	Depth of locking setscrews		Table I		Manual/Gage
103.	Identification symbols match on oversize pins and holes where applicable		Table I	3.4	Visual
104.	Objective evidence available for silicone fluid in accordance with dwg 8857555			6.2	Visual
Minor:					
201.	Marking		Table II		Visual
202.	Workmanship		Table II	3.8	Visual/Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1.2	Firing Set MC-1253 and Timer, Sequential M32E1 with Plug, Test Assembly Before Assembly of Firing Set				8857469
					NEXT HIGHER ASSEMBLY 8857468
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Dial reads 999 to 000 inclusive or 000 to 001 inclusive		100%		Visual
2.	Seal on sequential timer, not broken or missing		100%		Visual
3.	Pin connector not damaged		100%		Visual
Minor:					
201.	Workmanship		Table II	3.8	Visual/Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8857469
4.3.2.1.3	Firing Set MC-1253 and Timer Sequential M32E1 with Plug, Test Assembly				NEXT HIGHER ASSEMBLY 8857468
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Key, sequential timer and firing set assembly surfaces are flush		100%		Visual/Manual
Major:					
101.	Gap correctly recorded between firing set subassembly and sequential timer		Table I		Gage
102.	Torque on setscrew		Table I		Gage
Minor:					
201.	Workmanship		Table II	3.8	Visual/Manual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1.4	Body ogive assembly (see dwg 8857468) covering a detail of dwg 9233668 or 9233664 as applicable before assembly of firing set MC-1253 and sequential timer with plug, test assembly (see dwg 8857469)	1 OF 1	8857468
			NEXT HIGHER ASSEMBLY: 9233668 or 9233664

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Previously recorded gap (see dwg 8857469) is decreased by .004 ± .002 inch		Table I		Gage
102.	Preformed packings (3) coated with silicone fluid		Table I		Visual
103.	No cracks or pin holes in mating surface between body and ogive		Table I		Visual
Minor:	None defined				

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1.5	Body ogive assembly		8857468
			NEXT HIGHER ASSEMBLY
			9233664 or 9233668

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	"X" on body assembly aligned with "X" on ogive assembly		100%		Visual
2.	Seal on front assembly not broken or missing		100%		Visual
3.	Dial on sequential timer reads 999 to 000 inclusive or 000 to 001 inclusive		100%		Visual
Major:					
101.	Depth of insert pin assemblies		Table I		Gage
102.	Identification symbols match on oversize pins and holes where applicable		Table I	3.4	Visual
103.	Sealing material present on joint between ogive and body		Table I		Visual
104.	Positive clutch half pin mated with groove in clutch half positive on sequential timer		Table I		Visual
105.	Objective evidence available for silicone fluid in accordance with dwg 8857468			6.2	Visual
Minor:					
201.	Workmanship		Table II	3.8	Visual/Manual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1.6	Insert assembly (when pin has been removed or hole reworked to eliminate damage)		8857555 & 8857468
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:	100% Inspection Before Assembly				
101.	Diameter of hole is straight and round within .0003 (see dwg 8857487)		100%	3.4	Gage
102.	Diameter of hole is not more than .3900 inch (see dwg 8857487)		100%	3.4	Gage
103.	Radius on bottom of hole, max (oversize holes only) (see dwg 8857487)		100%	3.4	Gage
104.	Diameter of pin will assure a .0008 to .0012 interferance fit (see dwg 8857487)		100%	3.4	Gage
105.	Length of pin (see dwg 8857477 or 8881785 as applicable)		100%		Gage
106.	Chamfer on pin, min (see dwgs 8857477 or 8881785 as applicable)		100%		Gage
107.	Radius present on hole (see dwg 8857487)		100%		Visual
	After Assembly				
108.	Depth of insert pin assemblies		100%		Gage
Minor:					
201.	Workmanship		Table II	3.8	Visual/Manual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1.7	Projectile assembly 155 millimeter shipping configuration				9233664 or 9233668
					NEXT HIGHER ASSEMBLY
					9233663 or 9233667
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Dial visible through window reads 999 to 000 inclusive or 000 to 001 inclusive		100%		Visual
2.	Seal on test plug not broken or missing		100%		Visual
3.	No visible scratch, nick or gouge on base, rear body, center body, or forward body not previously repaired		100%	3.5	Visual/Manual/Gage
Major:					
101.	Packings present in retainer, test plug before assembly to projectile		Table I		Visual
102.	Retainer, test plug properly assembled		Table I		Visual/Manual
103.	Rotating band		100%	3.6	Visual/Manual/Gage
Minor:					
201.	Marking		Table II		Visual
202.	Protective finish missing including bare spot		Table II		Visual
203.	Serialization		Table II	3.7	Visual
204.	Workmanship		Table II	3.8	Visual/Manual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1.8	Container, shipping and storage: M467 packaging and marking for projectile M454 A4 or A6 (see dwg 9241967) prior to closing cover.	1 OF 3	9241967
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	No visible scratch, nick or gouge on base rear body, center or forward body caused by assembly in container		100%	3.5	Visual/Manual/Gage
Major:	100% Inspection				
101.	Rotating band before assembly of clamp		100%	3.6	Visual/Manual/Gage
102.	Torque on washer-screws on retaining clamp assemblies		100%		Gage
103.	Handling device properly assembled		100%		Visual
104.	Web strapping properly secured with tape		100%		Visual
105.	Base of projectile seated against back plate of rear cradle		100%		Visual
106.	Talcum powder present on rubber pads and cushions (note: observe before packing projectile)		100%		Visual
107.	Fuse present		100%		Visual.
108.	Packing list and maintenance equipment log properly packed		100%		Visual
109.	Desiccant present		100%		Visual
110.	Seal flanges clean		100%		Visual
111.	No nicks, cuts or other damage in gasket		100%		Visual
112.	Web straps and buckles not loose, missing or broken		100%		Visual
113.	Rubber strips not loose missing or broken		100%		Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.3.2.1.8	Container, shipping and storage: M467 packaging and marking for projectile M454 A4 or A6 (see dwg 9241967) prior to closing cover.		9241967
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
114.	Two plastic covered chains not loose missing or broken		100%		Visual
115.	Rubber padding between projectile and cradle assembly and clamp assembly not loose, missing or broken		100%		Visual
116.	Cushioning material top and bottom, not loose, missing or broken		100%		Visual
117.	Resilient mounts not loose, missing or broken		100%		Visual
118.	No dents more than one inch deep on T361 container		100%		Gage
119.	No holes, cracks, or separations in T361 container		100%		Visual
120.	Male contacts of same diameter appear to be within 1/16 inch in height relative to adjacent pins on connector of T361 container		100%		Visual
121.	Three or more contacts do not appear to be bent (one or more contacts may be straightened providing the bends are singular in nature) in connector on T361 container		100%		Visual
122.	No contacts missing, as evidenced by empty holes in insert in connector on T361 container		100%		Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1.8	Container, shipping and storage: M467 packaging and marking for projectile M454 A4 or A6 (see dwg 9241967) prior to closing cover.	3 OF 3	9241967
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
123.	No foreign material on contacts or inserts on connector on T361 container		100%		Visual
124.	No imperfections of conenctor on T361 container which affect connection		100%		Visual
125.	No stripped threads on connector on T361 container		100%		Visual
126.	No cracked coupling rings, clamps, shells, adapters or inserts on connector T361 container		100%		Visual
127.	No missing portions of shells, clamps or adapters or connector on T361 container		100%		Visual
Minor:					
201.	Workmanship		Table II	3.8	Visual/Manual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1.9	Container, shipping and storage: M467 packaging and marking for projectile: M454 A4 or A6 (see dwg 9241967) after closing cover.		9241967
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	No latch loose or secured improperly		Table I		Manual/Visual
102.	No car seal missing or secured improperly		Table I		Manual/Visual
103.	Marking		Table I		Visual
104.	Humidity indication more than 30%		Table I		Visual
105.	No dents more than one inch deep		Table I		Gage
106.	No holes, cracks or separations		Table I		Visual
107.	No hoisting graples, loose, missing or broken		Table I		Visual
108.	No handles, loose, missing or broken		Table I		Visual
109.	No runners, loose, missing or broken		Table I		Visual
110.	No vent assembly, loose, missing or broken		Table I		Visual
111.	No humidity indicator assembly, loose, missing or broken		Table I		Visual
112.	No valve assembly, loose missing or broken		Table I		Visual
113.	No tie down rings, missing or broken		Table I		Visual
Minor:					
201.	No protective finish damaged including bare spot		Table II		Visual
202.	Workmanship		Table II	3.8	Visual/Manual

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4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test	Test Classification
a. Circuit resistance	Critical
b. Insulation resistance	Critical

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance (see dwg 8857469). Each assembly shall be tested as specified in 4.4.1. If any assembly fails to comply with drawing requirements the assembly shall be rejected and removed from the lot.

4.3.3.2 Open circuit resistance (see dwg 8857469). Each assembly shall be tested as specified in 4.4.2. If any assembly fails to comply with drawing requirements, the assembly shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-9233668. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the equipment used.

4.4.1 Circuit resistance (see 4.3.3.1). The circuit resistance shall be measured with an approved suitable low resistance measuring device.

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4.4.2 Open circuit resistance (see 4.3.3.2). Insulation resistance shall be measured with an approved measuring device at 500 + 50 volts DC with current limited to 5 milliamperes.

4.4.3 Scratches, nicks or gouges (see 3.5). Any visible scratch, nick, or gouge shall be examined and repaired if necessary with the following procedures:

1. Remove paint with paint remover if necessary.
2. Clean area of surface of the imperfection with a brush or air hose.
3. If the visible scratch cannot be felt with a thumbnail on a 63 finish surface, ignore it (see 3.5).
4. If the imperfection can be felt with a thumbnail make an impression with "cast stone" (or grey stone) and determine the depth of the imperfection by a comparatator type device or any other accurate measuring device (approved by US Army Armament Research and Development Command, Dover, New Jersey, 07801 ATTN: DRDAR-QAN-Q), which does not contact the sample.
5. If the imperfection is deeper than 0.008 inch deep, reject the component (see 3.5).
6. If the imperfection is located near a fillet, chamfer, thread, or radii (see 3.5) and deeper than .005 inch on the base, rear body, and forward body or is greater than .003 inch deep on the center body, reject the component.
7. If the imperfection is felt by the thumbnail test but is less than previously defined depths (depending on location) the imperfection shall be repaired as follows:
   a. Use 240 abrasive grit emery cloth or finer grit size, with a suitable holder such as a wooden dowel or a soft metal dowel (aluminum, brass etc.) of no greater diameter than 1/2 inch, and rub lightly until the imperfection appears to be removed. Maximum width of repaired imperfection shall be 0.1 inch and maximum accumulated length of repaired imperfection on center body shall be 20 inches (see 3.5).
   b. Take a cast stone impression and proceed to measure area as described in step 4.
   c. If the imperfection is still present, continue steps a and b until it is removed.
   d. Accept or reject the component as specified in 3.5.
   e. Retain the first and last cast stone impressions for inspection records until projectile is shipped to stockpile.

4.4.4 Damage to rotating band (see 3.6). An impression with cast stone (or grey stone) shall be taken of each scratch, dent, nick, or gouge to determine the depth of the imperfection by a comparator type device or any other accurate measuring device approved by US Army Armament Research and Development Command, DRDAR-QAN-Q.

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Any other method devised by the contractor to measure the depth of imperfections shall be approved by US Army Armament Research and Development Command, DRDAR-QAN-Q. For allowable depth of the imperfection see Figure I. In addition, the rotating band shall be visually inspected for any crack or foreign matter embedded in the rotating band.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The projectile shall be packaged in accordance with dwg 9241967 using the specified moisture barrier.

5.1.2 Level C. The projectile shall be packaged in accordance with dwg 9241967 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The projectile shall be packed in accordance with dwg 9241967.

5.3 Marking. Marking shall be as specified in accordance with dwg 9214967.

5.4 Data cards. ERDA will supply serial numbers and record of assembly of each assembly produced to DRDAR-QAN-R, US Army Armament Research and Development Command, Dover, New Jersey 07801. US Army Armament Research and Development Command DRDAR-QAN-R shall assign a serial number compatible with the computing system and record on the data card all deviations which are accepted on the projectile.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, US Army Armament Research and Development Command, Dover, New Jersey 07801, ATTN: DRDAR-QAN-Q.

6.4 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings and specifications

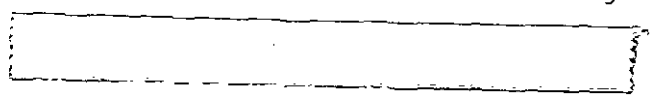
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prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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Preparing Activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-50286A Projectile, 155 Millimeter: M454, With Accessories DOD Comp.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? *PLEASE EXPLAIN BELOW.*

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